

PORVOO HARABACKA, Finland

WMO: 027590

Lat: 60.3917N

Lon: 25.6072E

Elev: 24

StdP: 101.04

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-18.6	-24.3	0.4	-21.8	-21.1	0.6	-18.5	9.8	-3.4	8.5	-3.4	2.3	0	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	27.5	19.2	25.5	18.1	23.8	17.1	20.6	25.3	19.3	23.8	18.1	22.4	4.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.8	23.3	17.6	12.6	21.6	16.4	11.7	20.0	59.5	25.3	55.1	24.0	51.3	22.5	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.5	DB	-25.2	29.5	3.9	2.2	-28.0	31.1	-30.2	32.3	-32.4	33.5	-35.2	35.1	(4)
(5)			WB	-24.3	21.3	5.2	1.3	-28.1	22.3	-31.1	23.0	-34.0	23.8	-37.8	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-6.1	-4.4	-1.3	4.3	11.0	14.8	18.0	16.3	12.0	5.6	2.1	-1.6	(6)
(7)		DBStd	9.08	6.43	6.10	4.32	3.26	4.12	3.22	3.16	2.80	3.12	3.55	4.18	5.34	(7)
(8)		HDD10.0	2221	498	404	351	172	36	3	0	1	17	142	237	360	(8)
(9)		HDD18.3	4581	757	637	609	420	230	114	46	76	191	396	487	618	(9)
(10)		CDD10.0	737	0	0	0	2	67	147	247	194	76	4	0	0	(10)
(11)		CDD18.3	56	0	0	0	0	2	7	35	12	0	0	0	0	(11)
(12)		CDH23.3	468	0	0	0	0	27	71	289	80	0	0	0	0	(12)
(13)		CDH26.7	80	0	0	0	0	2	8	57	13	0	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.4	3.4	3.5	3.2	3.3	3.5	3.1	3.0	3.0	3.3	3.4	3.7	(14)
(15)	Precipitation	PrecAvg	679	59	44	35	34	40	66	64	74	63	71	75	58	(15)
(16)		PrecMax	882	117	108	66	71	73	134	160	151	161	175	195	147	(16)
(17)		PrecMin	550	9	3	6	6	4	19	0	5	12	19	12	15	(17)
(18)		PrecStd	93	25	26	15	19	21	32	38	35	39	35	36	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	6.1	10.1	19.0	26.3	27.9	30.4	28.5	21.3	14.3	10.6	8.4	(19)
(20)			MCWB	4.0	3.2	5.6	10.6	16.0	18.4	19.8	19.5	16.4	11.4	9.4	7.5	(20)
(21)		2%	DB	3.8	3.8	7.4	16.0	23.5	25.3	28.2	25.4	19.5	12.8	9.1	6.5	(21)
(22)			MCWB	3.3	2.6	3.8	9.2	14.4	17.0	19.8	18.7	15.6	11.0	8.5	5.6	(22)
(23)		5%	DB	2.8	2.7	5.4	13.1	21.5	23.5	26.5	23.4	18.3	11.8	8.0	5.3	(23)
(24)			MCWB	2.4	2.0	2.9	7.4	13.4	15.8	19.2	17.4	14.6	10.6	7.3	4.6	(24)
(25)		10%	DB	1.6	2.0	3.6	10.6	19.2	21.6	24.7	21.8	17.0	10.8	7.2	4.3	(25)
(26)			MCWB	1.2	1.4	2.0	5.9	12.3	15.1	18.4	16.5	14.3	9.6	6.6	3.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.6	6.1	11.0	17.8	19.9	22.0	21.6	17.3	12.7	9.6	7.9	(27)
(28)			MCDWB	4.5	4.7	8.8	17.0	23.9	24.8	27.1	26.4	19.9	13.7	10.5	8.3	(28)
(29)		2%	WB	3.3	2.8	4.3	9.5	15.9	18.3	21.1	19.7	16.4	11.7	8.5	5.8	(29)
(30)			MCDWB	3.7	3.6	6.5	14.8	20.4	23.4	25.9	23.1	18.4	12.3	9.0	6.3	(30)
(31)		5%	WB	2.5	2.2	3.2	8.1	14.4	16.9	20.2	18.4	15.7	10.8	7.4	4.7	(31)
(32)			MCDWB	2.8	2.8	4.7	11.9	19.5	21.9	24.9	22.0	17.5	11.5	7.8	5.2	(32)
(33)		10%	WB	1.3	1.5	2.3	6.7	12.9	15.7	19.0	17.4	14.8	9.8	6.7	3.7	(33)
(34)			MCDWB	1.6	1.9	3.3	9.9	18.3	20.2	23.6	20.7	16.5	10.6	7.1	4.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	5.1	6.7	8.7	10.2	8.9	9.0	8.5	7.4	5.7	4.0	4.2	(35)
(36)			MCDBR	3.3	4.2	8.5	14.0	13.7	12.1	11.5	11.4	9.3	6.7	4.2	4.3	(36)
(37)			MCWBR	3.2	3.5	5.6	7.7	6.3	5.4	4.7	5.7	5.8	5.2	4.0	4.1	(37)
(38)		5% WB	MCDWR	3.4	3.9	7.2	11.7	11.8	10.8	10.2	9.3	7.9	5.3	4.1	4.0	(38)
(39)			MCWBR	3.4	3.4	5.2	6.9	6.1	5.6	4.7	5.1	5.3	4.8	4.0	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.276	0.310	0.334	0.364	0.362	0.359	0.380	0.372	0.346	0.332	0.303	0.383	(40)
(41)		taud		2.133	2.182	2.243	2.334	2.406	2.455	2.414	2.434	2.498	2.391	2.209	2.604	(41)
(42)		Ebn at Noon		468	667	784	832	864	872	840	818	780	651	432	326	(42)
(43)		Edh at Noon		42	72	94	103	104	101	103	93	74	60	40	28	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.25	0.82	2.31	3.84	5.20	5.41	5.25	4.04	2.46	1.07	0.30	0.14	(44)
(45)		RadStd		0.04	0.17	0.30	0.40	0.56	0.42	0.48	0.42	0.32	0.17	0.06	0.02	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	-142	N/A	N/A	N/A

Nomenclature: See separate page